Swiss TPH

Scientific Figure and Illustration

A picture is worth a thousand words

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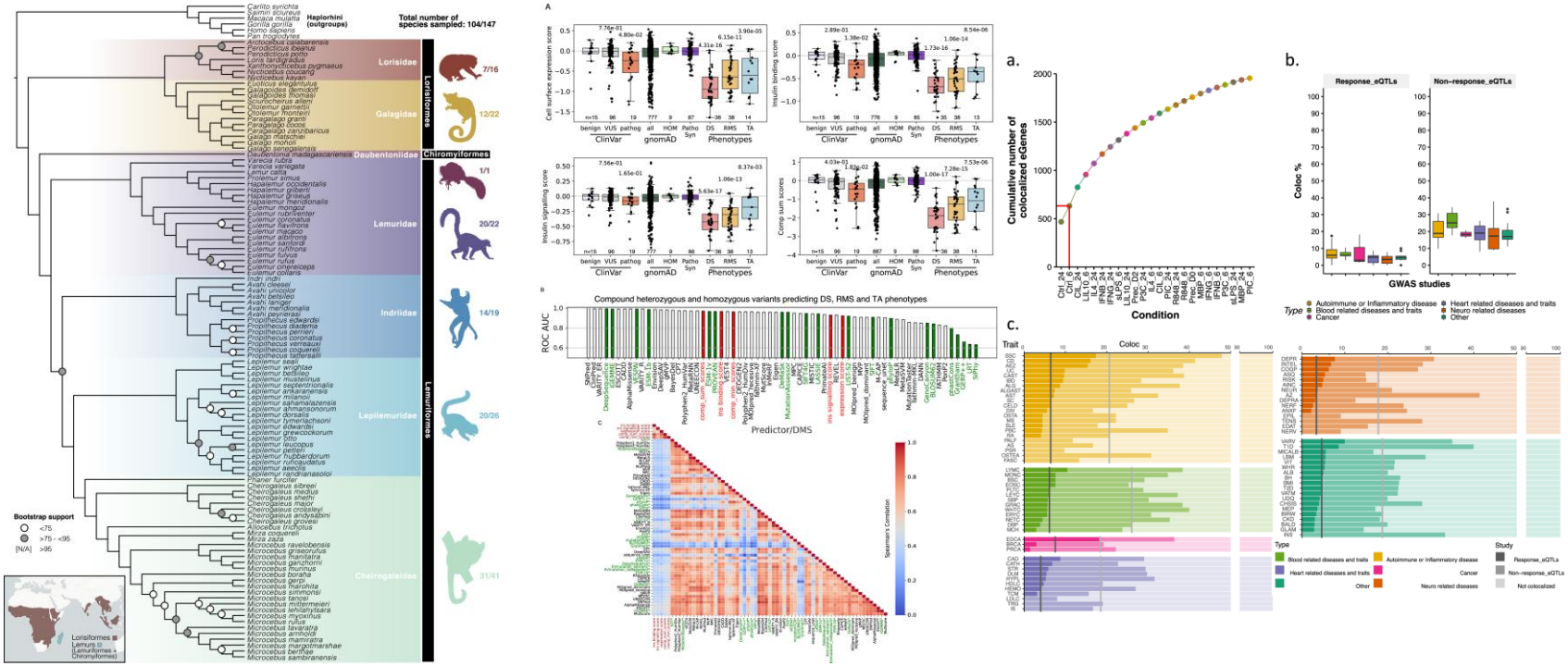
Research IT

How to make high quality, publication level figures?



Strong visuals secure grants/publications!

Figure source: Kathern et al, Vahid et al, Nikolaos et al

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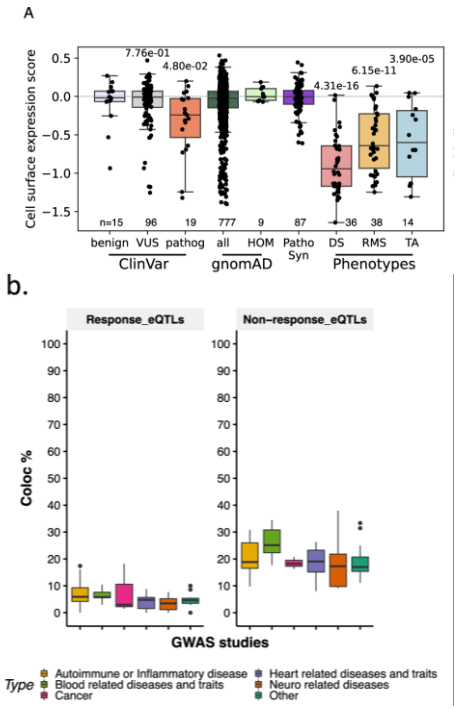
Color selection

Color palette instead of RGB



- Adobe color palette
- Color Hunt
- Swiss TPH color palette
- Color Space

Villains:

- #D1495B
- #E07A5F
- #F2CC8F
- #F4A261
- #D17C32

Protectors:

- #3D84A8
- #4EA8DE
- #6C91C2
- #3A5A40
- #274C77

Base palette:

- #bf3227
- #468ab2
- #b5c751
- #edcd64
- #868686

Poor contrast:

- #b84b43
- #6a96ab
- #aab77a
- #c8b57c
- #8a8a8a

Low saturation:

- #d5a6a2
- #9ab3c1
- #ccd7a8
- #e9ddb2
- #bfbfbf

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Create figures elegantly

Websites and resources for figures

- Biorender

A huge collection of health-related graphic materials and templates.
(Please email us if interested.)

- Freepic

A free all-type image resource library.

- Smart / Community Health Toolkit

Specialized in medical / health care vector resources.

Choosing the right graph

Understand your data

- **Categorical** ➡️ groups/labels/categories
- **Continuous** ➡️ numeric values

Focus on the story

- **Comparison**? 👉 bar chart, column chart, line chart, …
- **Distribution**? 👉 histogram, line/density histogram, box plot, …
- **Composition**? 👉 pie chart, stacked bar chart, stacked area chart …
- **Relationship**? 👉 scattered plot, bubble chart, heatmap, …

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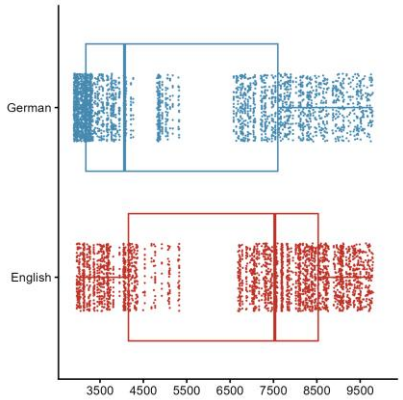
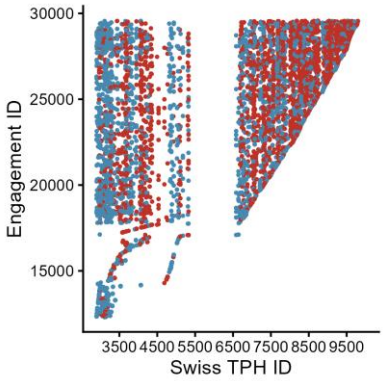
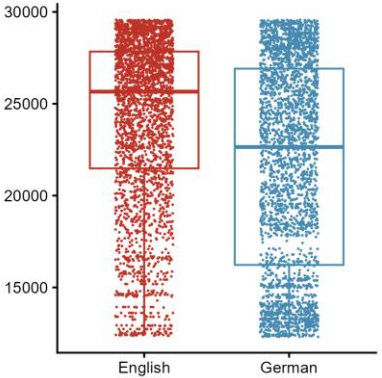
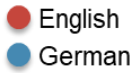
Figure assembly

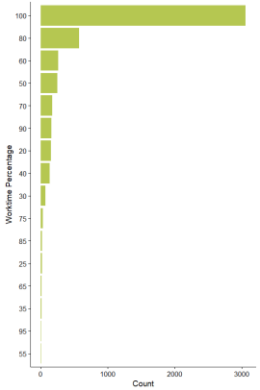
A bad example

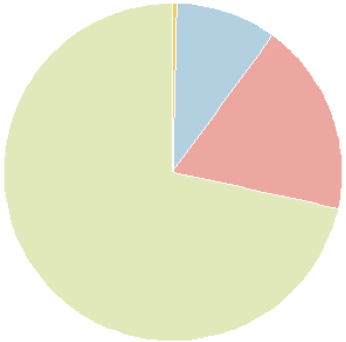


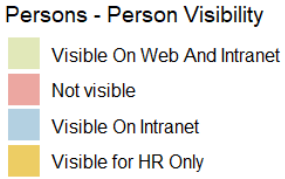










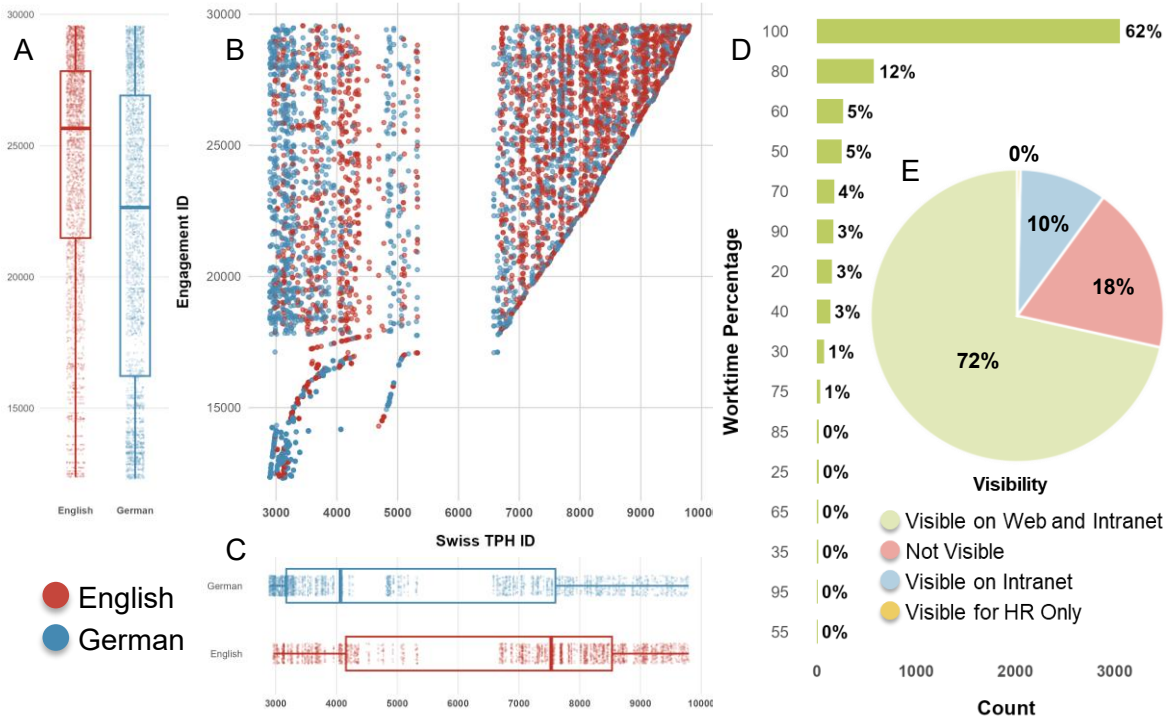


- Nothing is aligned
- Did not make full use of the canvas
- Not HD
- Font size too small
- No details
- ……

Last Step: Polishing your figure

- Reading order
- Align things
- Rank the data
- Adjust the canvas size
- Export as PDF

……



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Design Figures Strategically

Smart figure preparation – for publications

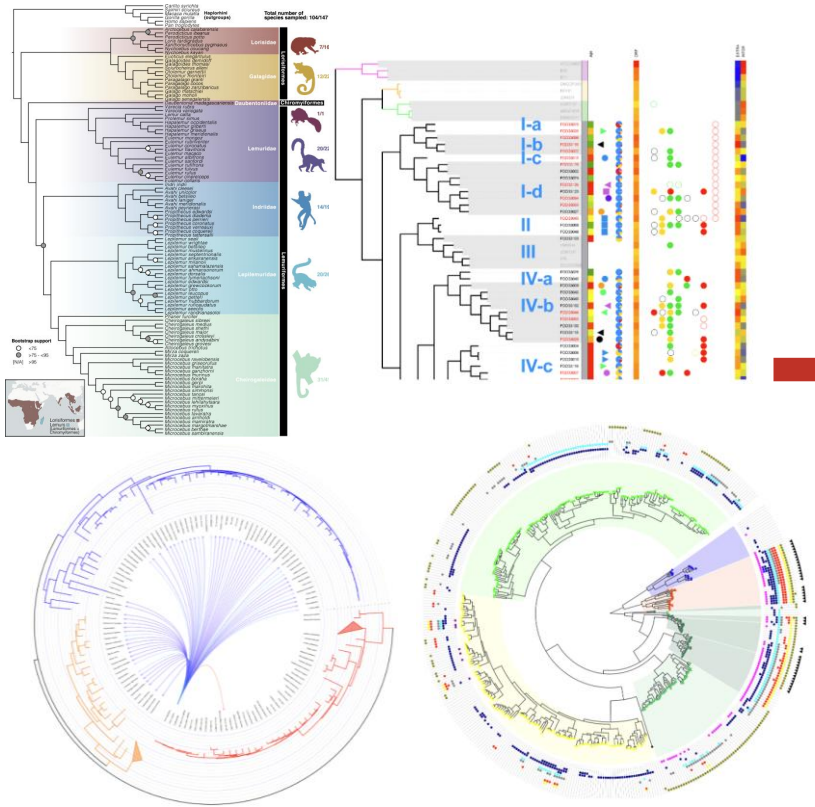
- Present **all the key findings** in your figures.
- Check the journal's **figure quota** before making the figures.
- Organize and **plan your figures** before drafting the manuscript.
- Don't jump in blindly - start by studying the figures in the **leading papers** in your field.
- Different figure for different audience.

- Practice makes perfect!

For discussion

- Which visualization tools or software do you recommend (and why)?
- Have you used AI tools (e.g., ChatGPT, Gemini, Claude) to help generate or improve figures?
- Any favorite resources or tutorials for improving figure design?
- How do you balance aesthetics vs. scientific accuracy?
- What are the biggest challenges when making figures (e.g., colors, layout, annotations, storytelling)?

Phylogenetic tree - iTol



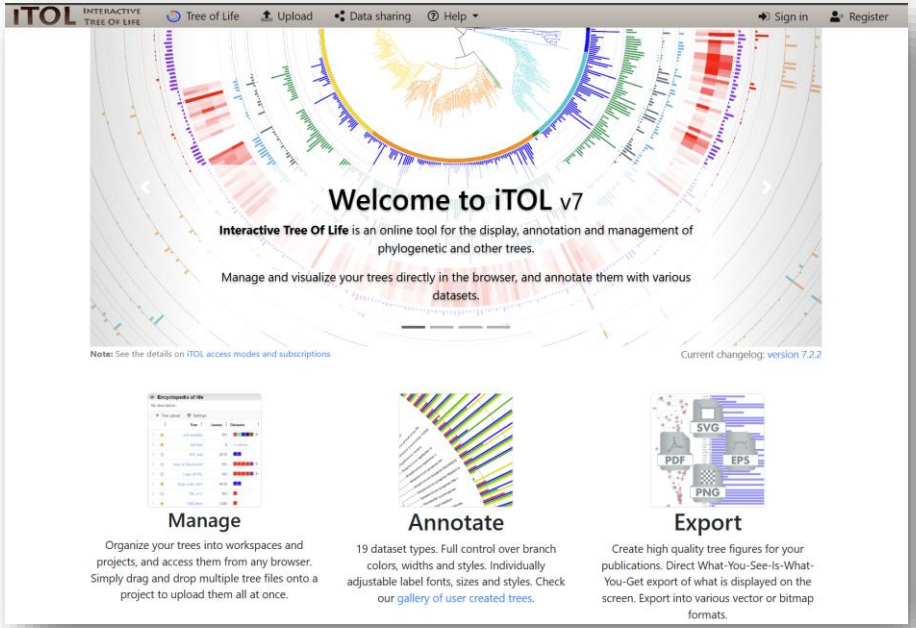


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Figure source: Kathern et al, Vahid et al, Nikolaos et al

Research-IT resources

Research-IT Documentation

- License available
- Troubleshooting tricks
- Quick start guides

SwissTPH Template (slides/Word/Quarto …)

Research IT Intranet

Feel free to contact us at research-it@swisstph.ch